TabuLLM: Comprehensive Augmentation Audit

1. Executive Summary

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference.

Metric	Consolidated Value
Avg Fidelity	85.2
Pipeline Status	Review

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Dataset Audit: None

1. Dataset Portrait

Attribute	Value
filename	sarcoma.csv
rows	500
cols	15
missing	0
memory	64 KB
targets	Target_Class

2. Generation Strategy & LLM Config

3. Statistical Significance & Data Drift

Aggregated across all k-folds. Drift defined as KS-Test p-value < 0.05.

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	45.2	45.8 ± 0.5	+1.2% ± 0.2	0.845 ± 0.05	Pass
Size	12.1	11.9 ± 0.8	-1.5% ± 0.6	0.912 ± 0.03	Pass

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category & Detail	Aggregated Score
Synthetic Fidelity (JSD) (Overall distri	85.20 ± 2.15
Correlation Stability (Multivariate rela	92.10 ± 1.50
Distribution Consistency (Median feature	84.50 ± 3.20
Entropy Preservation (Information divers	98.10 ± 0.50
Mean Z Score Drift (Standardized distrib	0.12 ± 0.05
Categorical Fidelity (Category distribut	95.00 ± 1.20
Privacy Risk (Synthetic rows too close t	2 ± 1
Data Density (Non-missing cells in augme	100.0 ± 0.0
Coverage Score (Synthetic coverage of or	88 ± 4
Feature Accuracy (Mean feature similarit	91 ± 2
Novelty Score (Distance ratio to nearest	0.85 ± 0.10

5. Downstream ML Utility Results

Model / Metric	Aggregated Performance (Mean ± Std)

LOGISTIC_REGRESSION (ACCURACY)	0.852 ± 0.02
LOGISTIC_REGRESSION (F1)	0.841 ± 0.03
LOGISTIC_REGRESSION (RMSE)	3.904 ± 0.77

Appendix: Multi-Fold Data Trace

Sample rows from original and each generated fold for audit trailing.

Baseline: Original Data Sample

Audit Trace: Fold 1

A	B
1	3
2	4